

Hamiltonian di Kuantum Materials

Bagian 1- Hamiltonian Fundamental

1. Schrodinger

Persamaan Schrodinger dapat kita tuliskan sebagai

$$-\frac{\hbar^2}{2m} \nabla^2 \Psi(\vec{r}, t) + V(r, t) \Psi(\vec{r}, t) = i\hbar \frac{\partial \Psi(\vec{r}, t)}{\partial t} \quad (1.1)$$

Jika terdapat medan magnet

$$\hat{H} = \frac{1}{2m} [\vec{p}^2 + e\vec{A}(\vec{r})]^2 + V(r) \quad (1.2)$$

2. Pauli Hamiltonian

Untuk spin-1/2

$$\hat{H} = \frac{1}{2m} [\vec{p}^2 + e\vec{A}]^2 - e\phi - \frac{e\hbar}{2m} \vec{\sigma} \cdot \vec{B} \quad (1.3)$$

3. Hamiltonian Dirac

$$\hat{H} = c\vec{\alpha} \cdot (\vec{p} + e\vec{A}) + \beta mc^2 - e\phi \quad (1.4)$$

4. Kohn-Sham Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + V_{ext} + V_H + V_{xc} \quad (1.5)$$

jika terdapat tambahan suku Coupling

$$\hat{H} = H_{KS} + \frac{\hbar}{4m^2 c^2} (\nabla V) \times \vec{p} \cdot \vec{\sigma} \quad (1.6)$$

5. Hamiltonian efektif massa

Untuk pita konduksi

$$H_c = E_c + \frac{\hbar^2 k^2}{2m^*} \quad (1.7)$$

Untuk bagian yang anisotropi, kita tinggal pecahkan saa persamaan (1.7)

6. Generic k.p

Ekspansi di sekitar $\mathbf{k}_o$

$$H(\mathbf{k}_o + \mathbf{q}) = H_o + \sum_i q_i \left. \frac{\partial H}{\partial k_i} \right|_{\mathbf{k}_o} + \frac{1}{2} \sum_{i,j} q_i q_j \left. \frac{\partial H}{\partial k_i} \frac{\partial H}{\partial k_j} \right|_{\mathbf{k}_o} + \dots \quad (1.8)$$

Dengan $\mathbf{q} = \mathbf{k} - \mathbf{k}_o$

Bagian 2 (Tight Binding)

7. Single-band Tight binding

$$H = \sum_i \varepsilon_i c_i^\dagger c_i - \sum_{\langle ij \rangle} t_{ij} (c_i^\dagger c_j + c_j^\dagger c_i) \quad (1.9)$$

8. Rantai 1D

$$H = \varepsilon_o \sum_i c_i^\dagger c_i - t \sum_i (c_i^\dagger c_{i+1} + h.c) \quad (1.10)$$

9. Square lattice

$$H(k) = \varepsilon_o - 2t \left[\cos(k_x a) + \cos(k_y a) \right] \quad (1.11)$$

Dengan tinjauan NNN

$$H(k) = \varepsilon_o - 2t_1 \left[\cos(k_x a) + \cos(k_y a) \right] - 4t_2 \left[\cos(k_x a) + \cos(k_y a) \right] \quad (1.12)$$

10. Cubic Lattice

$$H(k) = \varepsilon_o - 2t \left[\cos(k_x a) + \cos(k_y a) + \cos(k_z a) \right] \quad (1.13)$$

11. Multi orbital TB

$$\sum_{i\alpha} \varepsilon_\alpha c_{i\alpha}^\dagger c_{i\alpha} + \sum_{ij} \sum_{\alpha\beta} t_{i\alpha,j\beta} c_{i\alpha}^\dagger c_{j\beta} \quad (1.14)$$

12. Slater-Koster

$$t_{sp_x = l(sp\sigma)} \\ t_{p_x p_x} = l^2(pp\sigma) + (1-l^2)(pp\pi) \quad (1.15)$$

$$t_{p_x p_y} = lm[(pp\sigma) - (pp\pi)]$$

$$H_{SK} = \sum_\alpha \varepsilon_\alpha |\alpha\rangle \langle \alpha| + \sum_{\alpha\beta} t_{\alpha\beta} |\alpha\rangle \langle \beta| \quad (1.16)$$

Bagian 3 Graphene dan 2D materials

13. Graphene

Basisnya A,B

$$H(k) = -t \begin{pmatrix} 0 & f(k) \\ f^*(k) & 0 \end{pmatrix} \quad (1.17)$$

Dengan $f(k) = \sum_j e^{i\vec{k} \cdot \vec{\delta}_j}$

14. Graphene+NNN

$$H = -t_1 \sum_{\langle ij \rangle} c_i^\dagger c_j - t_2 \sum_{\langle\langle ij \rangle\rangle} c_i^\dagger c_j + h.c \quad (1.18)$$

15. Low energy graphene Dirac

$$H_\tau = \hbar v_F (\tau q_x \sigma_x + q_y \sigma_y) \quad (1.19)$$

Dengan spin

$$H_\tau = \hbar v_F (\tau q_x \sigma_x + q_y \sigma_y) \otimes s_o \quad (1.20)$$

16. Gapped graphene

$$H_\tau = \hbar v_F (\tau q_x \sigma_x + q_y \sigma_y) + m \sigma_z \quad (1.21)$$

17. Kane Mele

$$H = -t_1 \sum_{\langle ij \rangle} c_i^\dagger c_j + i \lambda_{SO} \sum_{\langle\langle ij \rangle\rangle} v_{ij} c_i^\dagger s_z c_j + h.c \quad (1.22)$$

Dengan Rashba

$$H_R = i \lambda_R \sum_{\langle\langle ij \rangle\rangle} c_i^\dagger \left(\vec{s} \times \vec{d}_{ij} \right)_z c_j \quad (1.23)$$

$$H_{KM} = H_t + H_{ISO} + H_R \quad (1.24)$$

Bagian 4 Hamiltonian material topologis

18. SSH

$$H = - \sum_n \left[t_1 a_n^\dagger b_n + t_2 a_{n+1}^\dagger b_n + h.c \right] \quad (1.25)$$

Dalam notasi Bloch

$$H(k) = \begin{pmatrix} 0 & t_1 + t_2 e^{-ika} \\ t_1 + t_2 e^{ika} & 0 \end{pmatrix} \quad (1.26)$$

Atau

$$H(k) = d_x \sigma_x + d_y \sigma_y \quad (1.27)$$

Dengan

$$d_x = t_1 + t_2 \cos(ka)$$

$$d_y = t_2 \sin(ka)$$

19. Rice-Mele

$$H(k) = (t_1 + t_2 \cos ka) \sigma_x + t_2 \sin ka \sigma_y + \Delta \sigma_z \quad (1.28)$$

20. QWZ

$$H = \sin k_x \sigma_x + \sin k_y \sigma_y + (m + \cos k_x + \cos k_y) \sigma_z \quad (1.29)$$

21. Haldane

$$\begin{aligned} H(k) &= d_o(k)I + d_x(k)\sigma_x + d_y(k)\sigma_y + d_z(k)\sigma_z \\ NN \\ d_x - id_y &= -t_1 f(k) \\ NNNComplexhopping \\ d_o &= 2t_2 \cos \phi \sum_i \cos(k \cdot b_i) \\ d_z &= M - 2t_2 \sin \phi \sum_i \sin(k \cdot b_i) \end{aligned} \quad (1.30)$$

22. Generic Chern insulator

$$H(k) = d_o I + d(k) \cdot \sigma \quad (1.31)$$

23. BHZ

$$H_{\text{BHZ}} = \begin{pmatrix} h(\mathbf{k}) & 0 \\ 0 & h^*(-\mathbf{k}) \end{pmatrix}.$$

Dengan

$$h(\mathbf{k}) = \epsilon(k)I + Ak_x \sigma_x + Ak_y \sigma_y + M(k) \sigma_z$$

dan

$$\epsilon(k) = C - Dk^2,$$

$$M(k) = M - Bk^2.$$

24. Massive Dirac

$$H = v(k_x\sigma_x + k_y\sigma_y) + m\sigma_z.$$

Untuk 3D:

$$H = v(k_x\sigma_x + k_y\sigma_y + k_z\sigma_z) + m\sigma_0.$$

Bagian 5 Dirac/Weyl/SOC

25. Weyl

$$H_W = \chi\hbar v_F(k_x\sigma_x + k_y\sigma_y + k_z\sigma_z).$$

26. Tilted Weyl

$$H_W = \mathbf{w} \cdot \mathbf{k} I + \chi\hbar v_F \mathbf{k} \cdot \boldsymbol{\sigma}.$$

27. Anisotropic Weyl

$$H_W = \mathbf{w} \cdot \mathbf{k} I + \chi(v_x k_x \sigma_x + v_y k_y \sigma_y + v_z k_z \sigma_z).$$

28. Dirac Semimetals

$$H_D = v\tau_x(k_x\sigma_x + k_y\sigma_y + k_z\sigma_z).$$

29. Rashba

$$H_R = \frac{\hbar^2 k^2}{2m^*} I + \alpha_R(k_y\sigma_x - k_x\sigma_y).$$

30. Dresselhaus

Linear:

$$H_D = \beta(k_x\sigma_x - k_y\sigma_y).$$

Cubic:

$$H_D = \gamma[k_x(k_y^2 - k_z^2)\sigma_x + k_y(k_z^2 - k_x^2)\sigma_y + k_z(k_x^2 - k_y^2)\sigma_z].$$

31. Rashba+Dresselhaus

$$H = \frac{\hbar^2 k^2}{2m^*} + \alpha(k_y \sigma_x - k_x \sigma_y) + \beta(k_x \sigma_x - k_y \sigma_y).$$

32. Zeeman

$$H_Z = -\frac{g\mu_B}{2}(B_x \sigma_x + B_y \sigma_y + B_z \sigma_z).$$

33. Exchange

$$H_{\text{ex}} = -JM \cdot \sigma.$$

Jika magnetization unit vector:

$$H_{\text{ex}} = -\Delta_{\text{ex}} \hat{\mathbf{m}} \cdot \sigma.$$

34. Rashba+Zeeman

$$H = \frac{\hbar^2 k^2}{2m} + \alpha_R(k_y \sigma_x - k_x \sigma_y) + V_Z \sigma_z.$$

Bagian 6

35. Ising

$$H_{\text{Ising}} = -J \sum_{\langle ij \rangle} S_i^z S_j^z - h \sum_i S_i^z.$$

36. XY

$$H_{\text{XY}} = -J \sum_{\langle ij \rangle} (S_i^x S_j^x + S_i^y S_j^y).$$

37. Heisenberg

$$H_{\text{Heis}} = \sum_{\langle ij \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j.$$

38. XXZ

$$H_{\text{XXZ}} = \sum_{\langle ij \rangle} [J_{xy}(S_i^x S_j^x + S_i^y S_j^y) + J_z S_i^z S_j^z].$$

39. Kitaev Honeycomb

$$H_K = - \sum_{\langle ij \rangle_\gamma} J_\gamma S_i^\gamma S_j^\gamma.$$

dengan

$$\gamma = x, y, z.$$

40. Dzyaloshinski-Moriya

$$H_{\text{DMI}} = \sum_{\langle ij \rangle} \mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j).$$

41. Double Exchange

$$H = -t \sum_{\langle ij \rangle} (c_i^\dagger c_j + h.c.) - J_H \sum_i \mathbf{S}_i \cdot c_i^\dagger \boldsymbol{\sigma} c_i.$$

42. Stoner

$$H = \sum_{i\sigma} \epsilon_i n_{i\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}.$$

Dalam mean field:

$$H_{\text{MF}} = H_0 - \Delta_{\text{ex}} \hat{\mathbf{m}} \cdot \boldsymbol{\sigma}.$$

Bagian 7 : Material terkorelasi

43. Hubbard

$$H_{\text{Hubbard}} = -t \sum_{\langle ij \rangle, \sigma} (c_{i\sigma}^\dagger c_{j\sigma} + h.c.) + U \sum_i n_{i\uparrow} n_{i\downarrow}.$$

44. Extended Hubbard

$$H = H_{\text{Hubbard}} + V \sum_{\langle ij \rangle} n_i n_j.$$

45. T-J

$$H_{tJ} = -t \sum_{\langle ij \rangle \sigma} (\tilde{c}_{i\sigma}^\dagger \tilde{c}_{j\sigma} + h.c.) + J \sum_{\langle ij \rangle} \left(\mathbf{S}_i \cdot \mathbf{S}_j - \frac{1}{4} n_i n_j \right).$$

46. Anderson impurity

$$H = \sum_{k\sigma} \epsilon_k c_{k\sigma}^\dagger c_{k\sigma} + \epsilon_d \sum_{\sigma} n_{d\sigma} + U n_{d\uparrow} n_{d\downarrow} + \sum_{k\sigma} (V_k c_{k\sigma}^\dagger d_{\sigma} + h.c.).$$

47. Kondo

$$H_K = \sum_{k\sigma} \epsilon_k c_{k\sigma}^\dagger c_{k\sigma} + J_K \mathbf{S} \cdot \mathbf{s}(0).$$

dengan

$$\mathbf{s}(0) = \frac{1}{2} c_0^\dagger \boldsymbol{\sigma} c_0.$$

Bagian 8 : Superkonduktivitas

48. BCS

$$H_{\text{BCS}} = \sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \sum_{\mathbf{k}} [\Delta c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger + h.c.].$$

49. BdG

Dengan Nambu spinor:

$$\Psi_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{\mathbf{k}\downarrow} \\ c_{-\mathbf{k}\uparrow}^\dagger \\ c_{-\mathbf{k}\downarrow}^\dagger \end{pmatrix},$$

$$H_{\text{BdG}} = \begin{pmatrix} H_0(\mathbf{k}) & \Delta(\mathbf{k}) \\ \Delta^\dagger(\mathbf{k}) & -H_0^*(-\mathbf{k}) \end{pmatrix}.$$

50. S-wave BdG

$$H_{\text{BdG}} = \xi_k \tau_z + \Delta \tau_x.$$

Energi:

$$E_k = \pm \sqrt{\xi_k^2 + \Delta^2}.$$

51. Kitaev Chain

$$H = -\mu \sum_i c_i^\dagger c_i - t \sum_i (c_i^\dagger c_{i+1} + h.c.) + \Delta \sum_i (c_i c_{i+1} + h.c.).$$

Bloch BdG:

$$H(k) = (-\mu - 2t \cos k) \tau_z + 2\Delta \sin k \tau_y.$$

52. Rashba topological superconductor

$$H = \left(\frac{\hbar^2 k^2}{2m} - \mu \right) \tau_z + \alpha k \sigma_y \tau_z + V_Z \sigma_x + \Delta \tau_x.$$

Ini salah satu Hamiltonian penting untuk Majorana physics.

Part 9

53. Kane 8 band

Secara blok:

$$H_{8 \times 8} = \begin{pmatrix} H_{\Gamma_6} & H_{\Gamma_6 \Gamma_8} & H_{\Gamma_6 \Gamma_7} \\ H_{\Gamma_8 \Gamma_6} & H_{\Gamma_8} & H_{\Gamma_8 \Gamma_7} \\ H_{\Gamma_7 \Gamma_6} & H_{\Gamma_7 \Gamma_8} & H_{\Gamma_7} \end{pmatrix}.$$

Basis:

$$\{|\Gamma_6, \pm 1/2\rangle, |\Gamma_8, \pm 3/2\rangle, |\Gamma_8, \pm 1/2\rangle, |\Gamma_7, \pm 1/2\rangle\}.$$

54. Luttinger Hamiltonian

$$H_L = -\frac{\hbar^2}{2m_0} \left[\left(\gamma_1 + \frac{5}{2}\gamma_2 \right) k^2 - 2\gamma_2 (\mathbf{k} \cdot \mathbf{J})^2 \right].$$

Dengan $J = 3/2$.

55. BHZ sebagai k.p

$$H_{\text{BHZ}} = (C - Dk^2)I + A(k_x\sigma_x + k_y\sigma_y) + (M - Bk^2)\sigma_z.$$

Bagian 10 Moire

56. Generic bilayer continuum

$$H = \begin{pmatrix} H_1(-i\nabla) & T(\mathbf{r}) \\ T^\dagger(\mathbf{r}) & H_2(-i\nabla) \end{pmatrix}.$$

57. Bistritzer-MacDonald

$$H_{\text{BM}} = \begin{pmatrix} h_{-\theta/2}(-i\nabla) & T(\mathbf{r}) \\ T^\dagger(\mathbf{r}) & h_{+\theta/2}(-i\nabla) \end{pmatrix}.$$

Dengan

$$h_\theta(\mathbf{k}) = -\hbar v_F R_\theta \mathbf{k} \cdot (\sigma_x, \sigma_y).$$

Interlayer:

$$T(\mathbf{r}) = \sum_{j=1}^3 T_j e^{-i\mathbf{q}_j \cdot \mathbf{r}}.$$

Bagian 11-Electron-Phonon

58. Generic electron-phonon

$$H = H_e + H_{\text{ph}} + H_{e-ph}.$$

Dengan

$$H_e = \sum_{n\mathbf{k}} \epsilon_{n\mathbf{k}} c_{n\mathbf{k}}^\dagger c_{n\mathbf{k}},$$

$$H_{\text{ph}} = \sum_{\mathbf{q}\nu} \hbar\omega_{\mathbf{q}\nu} \left(b_{\mathbf{q}\nu}^\dagger b_{\mathbf{q}\nu} + \frac{1}{2} \right),$$

dan

$$H_{e-ph} = \sum_{mn\mathbf{k}\mathbf{q}\nu} g_{mn\nu}(\mathbf{k}, \mathbf{q}) c_{m,\mathbf{k}+\mathbf{q}}^\dagger c_{n,\mathbf{k}} (b_{\mathbf{q}\nu} + b_{-\mathbf{q}\nu}^\dagger).$$

59. Holstein

$$H_{\text{Holstein}} = -t \sum_{\langle ij \rangle} (c_i^\dagger c_j + h.c.) + \omega_0 \sum_i b_i^\dagger b_i + g \sum_i n_i (b_i + b_i^\dagger).$$

60. Frohlich

$$H_{\text{Fröhlich}} = \sum_k \epsilon_k c_k^\dagger c_k + \sum_q \hbar\omega_q b_q^\dagger b_q + \sum_{kq} M_q c_{k+q}^\dagger c_k (b_q + b_{-q}^\dagger).$$

61. SSH electron-phonon

$$H_{\text{SSH}} = - \sum_i [t_0 - \alpha(u_{i+1} - u_i)] (c_i^\dagger c_{i+1} + h.c.) + H_{\text{ph}}.$$

Bagian Xii Disorder

62. Anderson

$$H = -t \sum_{\langle ij \rangle} (c_i^\dagger c_j + h.c.) + \sum_i V_i n_i.$$

Dengan

$$V_i \in [-W/2, W/2].$$

63. Binary disorder

$$V_i = \begin{cases} +V_0, & i \in A \\ -V_0, & i \in B. \end{cases}$$

64. Magnetic Disorder

$$H_{\text{dis}} = \sum_i J_i \mathbf{S}_i \cdot \boldsymbol{\sigma}.$$

Bagian XIII Magnetic Field

65. Pierls substitution

$$t_{ij} \rightarrow t_{ij} \exp \left[\frac{ie}{\hbar} \int_{\mathbf{r}_i}^{\mathbf{r}_j} \mathbf{A} \cdot d\mathbf{l} \right].$$

Sehingga

$$H = - \sum_{\langle ij \rangle} t_{ij} e^{i\phi_{ij}} c_i^\dagger c_j + h.c.$$

66. Landau Hamiltonian

$$H = \frac{1}{2m} (\mathbf{p} + e\mathbf{A})^2.$$

Dengan

$$E_n = \hbar\omega_c \left(n + \frac{1}{2} \right).$$

67. Dirac Landau

Untuk graphene:

$$E_n = \text{sgn}(n) v_F \sqrt{2e\hbar B |n|}.$$

Bagian XIV Floquet

68. Time Periodic Hamiltonian

$$H(t + T) = H(t).$$

Ekspansi:

$$H(t) = \sum_m H_m e^{im\omega t}.$$

Floquet:

$$(H_F)_{mn} = H_{m-n} + m\hbar\omega\delta_{mn}.$$

69. Driven two-band model

$$H(t) = \mathbf{d} \left(\mathbf{k} + \frac{e}{\hbar} \mathbf{A}(t) \right) \cdot \boldsymbol{\sigma}.$$

Untuk circular polarization:

$$\mathbf{A}(t) = A_0(\cos \omega t, \sin \omega t).$$

Bagian XV Non Hermitian

70. Generic non-Hermitian

$$H \neq H^\dagger.$$

Misalnya:

$$H = \begin{pmatrix} \epsilon + i\gamma & t \\ t & \epsilon - i\gamma \end{pmatrix}.$$

Eigenvalue:

$$E_{\pm} = \epsilon \pm \sqrt{t^2 - \gamma^2}.$$

71. Non-hermitian SSH

$$H \neq H^\dagger.$$

Misalnya:

$$H = \begin{pmatrix} \epsilon + i\gamma & t \\ t & \epsilon - i\gamma \end{pmatrix}.$$

Eigenvalue:

$$E_{\pm} = \epsilon \pm \sqrt{t^2 - \gamma^2}.$$

Bagian 16: wannier

72. Wannier Hamiltonian

Wannier matrix element:

$$H_{mn}(\mathbf{R}) = \langle 0m | \hat{H} | \mathbf{R}n \rangle.$$

Bloch Hamiltonian:

$$H_{mn}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} H_{mn}(\mathbf{R}).$$

Jadi bentuknya identik dengan multi-orbital TB:

$$H(\mathbf{k}) = \sum_{\mathbf{R}} H(\mathbf{R}) e^{i\mathbf{k} \cdot \mathbf{R}}.$$

Bagian XVII: Light matter

73. Length Gauge

$$H(t) = H_0 + e\mathbf{E}(t) \cdot \mathbf{r}.$$

74. Velocity Gauge

$$\mathbf{p} \rightarrow \mathbf{p} + e\mathbf{A}(t)$$

sehingga

$$H(t) = H_0 \left(\mathbf{k} + \frac{e}{\hbar} \mathbf{A}(t) \right).$$

75. Dipole interaction

$$H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}(t)$$

dengan

$$\mathbf{d} = -e\mathbf{r}.$$

76. Generic nonlinear optical response

$$P_i = \epsilon_0 \left[\chi_{ij}^{(1)} E_j + \chi_{ijk}^{(2)} E_j E_k + \chi_{ijkl}^{(3)} E_j E_k E_l + \dots \right].$$

Bagian 18: quantum geometry

77. Berry Connection

$$A_i^{(n)} = i \langle u_n | \partial_{k_i} u_n \rangle.$$

78. Berry Curvature

$$\Omega_{ij}^{(n)} = \partial_{k_i} A_j - \partial_{k_j} A_i.$$

Equivalent:

$$\Omega_{ij}^{(n)} = -2 \text{Im} \langle \partial_i u_n | Q_n | \partial_j u_n \rangle$$

dengan

$$Q_n = 1 - |u_n\rangle \langle u_n|.$$

79. Quantum Metric

$$g_{ij}^{(n)} = \text{Re} \langle \partial_i u_n | Q_n | \partial_j u_n \rangle.$$

80. Quantum Geometric Tensor

$$Q_{ij}^{(n)} = \langle \partial_i u_n | Q_n | \partial_j u_n \rangle.$$

dan

$$Q_{ij} = g_{ij} - \frac{i}{2} \Omega_{ij}.$$

81. Two band QGT

Untuk

$$H = d_0 I + \mathbf{d} \cdot \boldsymbol{\sigma},$$

definisikan

$$\hat{\mathbf{d}} = \frac{\mathbf{d}}{|\mathbf{d}|}.$$

Maka

$$g_{ij} = \frac{1}{4} \partial_i \hat{\mathbf{d}} \cdot \partial_j \hat{\mathbf{d}}.$$

dan

$$\Omega_{ij} = -\frac{1}{2} \hat{\mathbf{d}} \cdot (\partial_i \hat{\mathbf{d}} \times \partial_j \hat{\mathbf{d}}).$$

82. Chern number

$$C_n = \frac{1}{2\pi} \int_{\text{BZ}} \Omega_{xy}^{(n)} d^2 k.$$

83. Zak phase

Untuk 1D:

$$\gamma_n = \oint_{\text{BZ}} A_k^{(n)} dk.$$

84. Orbital magnetic moment

$$\mathbf{m}_n(\mathbf{k}) = -\frac{e}{2\hbar} \text{Im} \langle \nabla_{\mathbf{k}} u_n | \times (H - E_n) | \nabla_{\mathbf{k}} u_n \rangle.$$

85. Berry curvature dipole

$$D_{ab} = - \sum_n \int_{\text{BZ}} \frac{\partial f_n}{\partial k_a} \Omega_{n,b} \frac{d^d k}{(2\pi)^d}.$$

Equivalent setelah integration by parts:

$$D_{ab} = \sum_n \int_{\text{BZ}} f_n \partial_{k_a} \Omega_{n,b} \frac{d^d k}{(2\pi)^d}.$$

Bagian 19 : Transport

86. Kubo Hamiltonian response

Generic conductivity:

$$\sigma_{ab}(\omega) = \frac{i\hbar}{V} \sum_{nm} \frac{f_n - f_m}{E_n - E_m} \frac{\langle n | j_a | m \rangle \langle m | j_b | n \rangle}{E_n - E_m + \hbar\omega + i\eta}.$$

87. Velocity operator

$$v_i = \frac{1}{\hbar} \frac{\partial H(\mathbf{k})}{\partial k_i}.$$

Ini sangat fundamental karena dari Hamiltonian langsung kita memperoleh

$$H \rightarrow v_i \rightarrow \sigma_{ij}.$$

88. Hall Conductivity

$$\sigma_{xy} = -\frac{e^2}{\hbar} \sum_n \int_{\text{BZ}} \frac{d^d k}{(2\pi)^d} f_n \Omega_{n,z}.$$

89. Anomalous Hall

$$\sigma_{xy}^{AHE} = -\frac{e^2}{\hbar} \sum_n \int f_n \Omega_{n,z}.$$

90. Thermoelectric

Secara umum:

$$L_{ij}^{(n)} = \int dE (E - \mu)^n \Sigma_{ij}(E) \left(-\frac{\partial f}{\partial E} \right).$$

Kemudian:

$$\sigma = L^{(0)}$$

$$S = -\frac{1}{eT} [L^{(0)}]^{-1} L^{(1)}$$

dan

$$\kappa_e = \frac{1}{e^2 T} \left[L^{(2)} - L^{(1)} [L^{(0)}]^{-1} L^{(1)} \right].$$

Bagian 20: quantum geometric engineering

91. Zeeman quantum geometry

$$H(\mathbf{k}, B) = H_0(\mathbf{k}) - \frac{g\mu_B}{2} \mathbf{B} \cdot \boldsymbol{\sigma}.$$

Kemudian:

$$g_{ij} = g_{ij}(\mathbf{k}, B)$$

dan

$$\Omega_{ij} = \Omega_{ij}(\mathbf{k}, B).$$

Kita dapat mendefinisikan:

$$\mathcal{G}_{ij}^{(B)} = \frac{\partial g_{ij}}{\partial B}$$

dan

$$\Omega_{ij}^{(B)} = \frac{\partial \Omega_{ij}}{\partial B}.$$

92. Strain quantum geometry

$$H(\mathbf{k}, \epsilon) = H_0(\mathbf{k}) + H_{\text{strain}}(\epsilon).$$

Kemudian

$$\mathcal{G}_{ij}^{(\epsilon)} = \frac{\partial g_{ij}}{\partial \epsilon}.$$

93. SOC quantum geometry

$$H = H_0 + \lambda_{\text{SOC}} H_{\text{SOC}}.$$

Sehingga

$$\frac{\partial g_{ij}}{\partial \lambda_{\text{SOC}}}$$

dan

$$\frac{\partial \Omega_{ij}}{\partial \lambda_{\text{SOC}}}.$$

94. Electric field quantum geometry

$$H(t) = H_0 + e\mathbf{E}(t) \cdot \mathbf{r}.$$

Secara nonlinear:

$$\mathcal{G}_{ij}^{(E)} = \frac{\partial g_{ij}}{\partial E}$$

dapat digunakan sebagai ukuran respons geometri terhadap medan listrik.